

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-1393
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

The technology choices below are aligned with the EduGenie project implementation: a web-based learning assistant with a Python backend and Gemini-powered AI features.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Builds the responsive web interface and provides the student-facing learning features such as questions, summaries and quizzes.
2	Backend / Server-Side	Python + FastAPI	Provides the server-side application and API endpoints, with Python supporting the EduGenie learning logic and FastAPI providing a lightweight web API framework.
3	Database / Data Storage	File-based / application data storage	The current EduGenie implementation does not require a dedicated relational database; application data can be handled by the project modules and temporary/request data.
4	Cloud / Hosting / Deployment	GitHub + local/hosted Python environment	GitHub stores the project source code and supports sharing/versioned development; the Python application can be run in a suitable hosting environment when deployed.
5	Version Control & CI/CD	Git + GitHub	Tracks source-code changes, supports commits and repository collaboration, and provides a central location for the EduGenie project.
6	Third-Party APIs / Other Tools	Google Gemini API / Gemini model	Provides the generative-AI capability used by EduGenie to create explanations, summaries, quizzes, answers and other learning assistance.

Team Members – Group 1

No.	Member Name	Group No.
1	BALAMURUGAN M	1
2	Aakash A	1
3	Akash S	1
4	Santhosh kumar R	1
5	Gunaseelan K	1